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RESEARCH ARTICLE

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Mechanics of bipedal and quadrupedal locomotion on dry and wet granular media

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Abstract

The effectiveness of exploration robots is contingent upon their capability and efficiency to locomote on contorted and multifaceted terrains. Traditional wheeled robots do not suffice at overcoming such terrains and complications they introduce. The proposed work explores the performance of WhegRunner, a whegged (i.e., wheel-legged) robotic platform used to examine the mechanics of running on granular media at different saturation levels. In particular, the effect of bipedal/quadrupedal gait, saturation level, stride length, and stride frequency on robot's forward body velocity and cost of transport (COT) was studied. To increment nominal stride length, the robot was fitted with different number of spoked whegs from 3 to 7. In addition, eight evenly spaced motor speeds ranging from 2.33 to 7.43 Hz were used to observe the effect of stride frequency. Similar trends were observed between the two gaits, with the quadrupedal gait showing better overall performance. On dry sand, wider strides were attained at reduced motor speeds with lower spoked whegs. However, on wet sand (15% and 30% saturation), wider strides were achieved with higher motor speeds and lower spoked whegs. As a result, lower spoked whegs accomplished higher body forward velocity and lower COT as saturation increased. The acquired knowledge elucidates underexplored mechanics of locomotion on granular media at different saturation levels. The WhegRunner can be used as a platform to obtain a greater insight on how to develop more proficient exploration robots, ones which can face complex and deformable terrains and mediums.

KEYWORDS
gait, soil mechanics, terrestrial robots, wheel–soil interaction

1 | INTRODUCTION

The interaction with natural media (e.g., sand, mud, gravel, leaves, and snow) and terrains (e.g., desert, swamp, steppe, forest, and tundra) restrict the mobility of many conventional robots. This is not to mention, the complications that arise from encountering with less understood media and terrains, such as those found upon the surface of the Moon and Mars. Despite the technological advancements of NASA's Mars Exploration Rover, Spirit, it still succumbed to slippage and sinkage, almost leading to the failure of the mission (Reina et al., 2006). Therefore, it becomes quite apparent that robotic systems have yet to transcend the obstacles produced by vehicle dynamic interactions with natural media. The understanding of such interactions is essential for the design and adaptation of effective ground exploration vehicles.

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Countless all-terrain vehicles and robots have been developed using limbless (e.g., snake) (Crespi et al., 2005; Crespi & Ijspeert, 2006; Ma et al., 2014; Marvi et al., 2014; S. Yu et al., 2011), limbed (Crespi et al., 2013; Floyd et al., 2008, 2006; Floyd & Sitti, 2008; Jun et al., 2013; Karakasiliotis et al., 2016; Kim et al., 2016; Lee et al., 2014; Low et al., 2007; Mazouchova et al., 2013; Park et al., 2009, 2010; Park & Sitti, 2009; Wang et al., 2006; Zhong et al., 2016), screw-propelled (Neumeyer & Jones, 1965; Osiński & Szykiedans, 2015; Thoesen, McBryan, et al., 2019; Thoesen et al., 2018; Thoesen, Ramirez, et al., 2019), wheeled (Sun & Ma, 2013; J. Yu et al., 2012, 2013), and whegged (i.e., legged-wheels) (Brown et al., 2013; Eich et al., 2008, 2009; Herbert et al., 2008; Lambrecht et al., 2005; Stager et al., 2015; Walker, 2011; Zarrouk et al., 2013; Zarrouk & Yehezkel, 2018) mechanisms for mobility.

Limbless (snake-like) robots can operate on diverse media, maintain excellent stability, and offer redundancy advantages. However, they are relatively slow, have limited payload capabilities, and can suffer from reliability issues due to their great number of degrees of freedom (DOF) (Crespi et al., 2005; Crespi & Ijspeert, 2006; Ma et al., 2014; Marvi et al., 2014; S. Yu et al., 2011). Screw-propelled vehicles provide great mobility over deformable terrains, but at the cost of high power consumption and low speeds (Osiński & Szykiedans, 2015). This is caused by the screw's large surface area contact. While wheeled robots can achieve high speeds with low power consumption and low DOF for navigation, they are limited in overcoming obstacles (Bruzzone & Quaglia, 2012). Highly complex mechanisms, such as multifunctional wheels or paddles, have a higher risk of failure due to the integration of numerous moving parts (Machado & Silva, 2006). Upon granular and deformable media, wheeled vehicles require awareness of their physical limitation and surrounding environment to avoid wheel slippage and sinkage, and at worst complete immobility. While wheeled vehicles have to greatly concern themselves with the interaction of natural substrates, legged locomotion provides an appealing alternative.

Another promising design for complex terrain mobility includes the use of compliant c-shaped legs or "whegs" in a hexapedal (cockroach-like) locomotion gait. The use of flexible legs for mechanical feedback can reduce design complexity and processing time required for electrical sensory feedback (Spagna et al., 2007). The interaction of c-shaped, flat, and reversed c-shaped legs with dry granular media showed that the stress experienced by the legs was dependent on the leg's depth, orientation, and direction of movement (Li et al., 2013). Through optimizing vehicle leg geometry and kinematics with granular media, granular solidification can be taken advantage of (Mazouchova et al., 2010). C-shaped and elliptical leg locomotion have been modeled on wet granular media (soil and mud) and validated with experimental studies (Liang et al., 2012; Ren et al., 2013; Xu et al., 2013; Zhang et al., 2013). While their efforts were of great contribution, some parameters were left unanalyzed. Specifically, how dry and saturated sand affect the robot's forward body velocity and cost of transport (COT) when traversed with bipedal and quadrupedal gaits, higher motor frequencies (i.e., above 0.5 Hz) are used and nominal stride lengths are varied. In addition, it is unclear how the robot's performance would change when removing support structures and allowing the robot to self-stabilize.

In this study, WhegRunner, a bipedal/quadrupedal locomotor is developed with whegs for ground mobility and an active tail for body pitch stability. The robotic platform is then employed to examine how forward body velocity and COT are affected when the following parameters are systematically altered: running gait (i.e., bipedal and quadrupedal), medium saturation (i.e., 0%, 15%, and 30% saturation), nominal stride length, and stride frequency. Through this study, greater insight was obtained on what it takes to design and develop effective and efficient ground exploration vehicles.

2 | METHODS

2.1 | Experimental setup

The 6.0 × 0.5 m^2 test bed (Figure 1) was used for robotic experiments. A 2 m segment of the test bed was partitioned and filled with 250–600 μm of sieved Sakrete Natural Play Sand, with medium height ranging between 55 and 60 mm, depending on saturation level (i.e., 0% and 30%). The sand was saturated by a uniform distribution of water over the surface and thorough mixing with an industrial hand mixer. A soil moisture sensor (GS1 manufactured by Decagon Devices) was used to measure the saturation level (i.e., volumetric

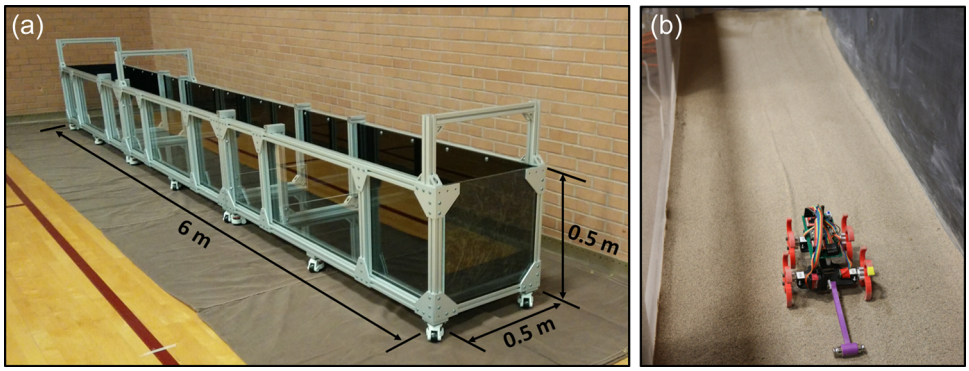


FIGURE 1 (a) Transition track, with (b) 2 m granular media experimental segment

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water content) of the sand at 10 equally spaced points in the track. The readings were averaged to obtain an overall saturation level of the test bed. Between trials, the test bed was manually scraped to create a relatively flat surface. In all, 0%, 15%, and 30% volumetric water content was used for experimentation. During preliminary experimentation, the decrease in medium saturation with respect to time was observed to be minimal. This was because experimentation was conducted under room conditions and no longer than 5 h sessions. In a day span, medium saturation was observed to drop from 15% to 12.5%. Therefore, before the beginning of each experimental session, the medium's saturation was measured and properly adjusted to the desired value. Saturation levels above 30% were not tested upon, for pockets of surfaced water became apparent. Therefore, 30% saturation was selected as the maximum, 0% as the minimum, and 15% as the medium. This allowed for three extreme saturation levels to be tested. The surface of the medium was manually compacted and leveled with a scraper before each experimental run. The experimental track was segmented into fourths. The first and last fourth of the track were for the start and finish of the robot, with the middle half for the recording of the run. While some runs reached a steady state much sooner (i.e., within the first fourth) than others, the middle half showed a steady state for all runs. Using a high-speed camera (SC1 Edgertronic), robot kinematics were captured from the side view.

2.2 | Medium characterization

The penetration stress of 0%, 15%, and 30% saturated medium was experimentally measured at different depths, taking inspiration from the procedure described in Stone et al. (2004). Saturation level of the medium was initially verified using a Decagon GS1 soil moisture sensor, then displaced into a 1-L glass beaker with 10.7 cm diameter and 14.5 cm height. The glass beaker was centered and mounted below a ceramic cylindrical rod with a contact diameter of 9 and 82 mm in length. The rod attached to a load cell (Interface 6A27A-F11) and a linear stage motor (Thorlabs MTS50-Z8) was vertically displaced into the medium with 2 mm/s velocity and 4 mm/s^2 acceleration. The selected velocity and acceleration were used to decrease the effect of particle inertia in dry sand. Five trials were conducted per 0%, 15%, and 30% saturation. Between each trial the medium was manually mixed and compacted (via manual tapping). Penetration stress (i.e., σ_z = force/contact area) has been plotted against vertical depth (z) of travel per each saturation level (Figure 2).

Medium penetration stress was observed to increase as indenter depth increased. The trend reflects more of a polynomial relationship, especially for saturated mediums, where a sudden increase in penetration stress is followed by a gradual curvilinear increase (i.e., convex). As saturation increased from 0% to 15% to 30%, penetration stress increased at a faster rate (e.g., higher stress obtained at lower penetration depths). In all, 0%, 15%, and 30% saturated mediums are classified as dry, pendular, and funicular, respectively, in the saturation spectrum. While dry granular media have little to no

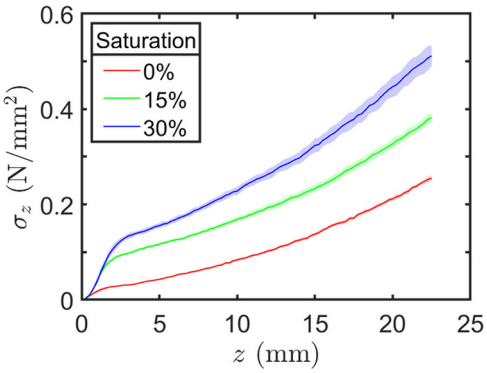


FIGURE 2 Penetration stress (σ_z) of 0%, 15%, and 30% saturated sieved (250–600 μm) Sakrete Natural Play Sand with respect to vertical depth (z) of travel by ceramic indenter with contact diameter of 9 mm. Five trials were performed per saturation

cohesion, in the pendular state, cohesive forces are formed through liquid capillary bridges between granular particles, yielding medium penetration stress. As the saturation content increases, air pockets between particles are filled with liquid (Mitarai & Nori, 2006). Therefore, with saturated mediums the forces that are in play are the medium's normal stress, air pressure (via air pockets), and liquid pressure (via capillary bridges). However, depending on the medium composition, medium stiffness (i.e., penetration force per depth) has been observed to increase (e.g., limestone and glass beads; Kristensen et al., 1985; Schubert et al., 1975) or decrease (e.g., dicalcium phosphate; Holm et al., 1985) with the increase of saturation. In our case, medium stiffness increased through the increase of saturation in the regime between 0% and 30%.

2.3 | Robot

WhegRunner (Figure 3) is a bipedal/quadrupedal robotic platform, which utilizes whegs (i.e., wheel-legs) for ground locomotion, and an active tail for counterbalancing the body pitch while running bipedally and inactive tail (tilted over the robot body) for the quadrupedal gait. The wheg design consists of half circle legs propagating from the central axis of the wheg hub. The number of legged spokes ranges from 3 to 7 (Figure 4) with each leg's outer (leading) diameter being 42.82 mm and leg thickness being 8 mm. The number of spokes correlated to different nominal stride lengths, where the lower number of spokes would provide wider strides and the higher number of spokes would provide shorter strides. The whegs and tail of the robot were comprised of 3D printed Acrylonitrile-Butadiene Styrene, providing a light, yet rigid structure. The wheg mass was preserved among all spoke designs at 11.11 ± 0.01 g by using different infill ratios for 3–7-spoke whegs (i.e., 74%, 53%, 37%, 26%, and 16%). The tail was 12.5 cm long with a total mass of 45 g, most of which was contributed at its end by a 40-g mass. Compliance in the wheg and tail design was avoided to

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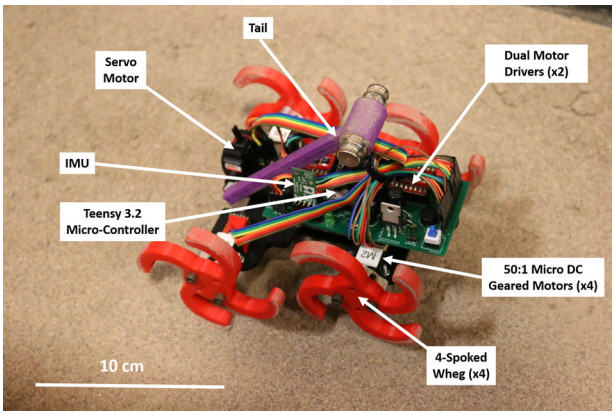


FIGURE 3 WhegRunner: Wheel-legged robot capable of both bipedal and quadrupedal motion. The body of the robot consists of an acrylic sheet base, a printed circuit board (PCB) housing a Teensy 3.2 controller, motor controllers, an inertial measurement unit (IMU) to determine body orientation, and a servo motor. Each multispoked wheg is actuated by a 50:1 Pololu DC geared motor and is modular in design, allowing multiple leg geometries to be mounted.

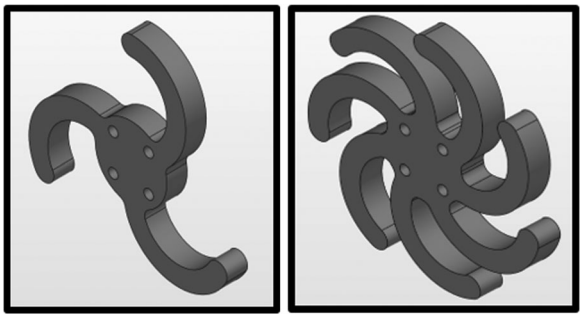


FIGURE 4 Wheel-leg or wheg spoke designs varied from three spokes (left) to seven spokes (right)

decrease the additional parameters that could affect body velocity. The axial separation between the front whegs (i.e., 65 mm) is less than the rear whegs (i.e., 95 mm). This was done to avoid front and rear whegs interference without increasing the body length. The robot body is a 2.4-mm thick acrylic sheet (laser cut to design), housing four 12 V Pololu 50:1 metal gearmotors with magnetic encoders, four aluminum motor hubs, printed circuit board with two SparkFun dual motor drivers, mini digital servo (HD-1810MG), a Teensy 3.2 microcontroller, and Pololu inertial measurement unit (AltIMU-10 v5). The total mass of the robot along with the weighted tail and averaged weight of all sets of whegs was 286 g.

2.4 | Robotic experiments

Bipedal and quadrupedal gait trials were separately experimented on closely packed dry (0% saturated) and wet (15% and 30% saturated) sand. To ensure alternating spoke contacts, the proportional–integral–derivative (PID) controller assured a phase shift between the robot's two rear whegs for the bipedal gait, and the syncing of diagonal whegs while at a phase shift with the opposing diagonal whegs for the quadrupedal (trot) gait. More specifically, the relative phases were controlled in the following manner: before each run, diagonal pairs were offset by a phase angle, which depends on the number of spokes per wheg. For example, a four-spoked wheg would have a phase angle of $360/(4 \times 2) = 45°$. The controller attempts to maintain this phase angle throughout the motion of the robot. The wheg position was determined by using feedback from magnetic encoders of each micromotor, each with a resolution of 1024 counts per revolution (cpr). This phase shift was maintained by monitoring the error between each pair of motors, and either speeding up or slowing down the motor pairs as necessary, using a PID controller. Through preliminary experimentation, the PID gains were tuned for each tested terrain (i.e., 0%, 15%, and 30% saturation). Similarly for bipedal motion, the rear whegs are out of phase and the above control scheme follows. For bipedal gait, the PID controller also corrected the tail angle to stabilize the robot's body angle as it advanced along the test bed. The robot's body angle was obtained from the attached IMU and used to experimentally tune the gains of the tail's PID controller per each saturation level. The tail was actuated by a single servo motor at the tail base. Occasional tail drag was unavoidable, especially in dry sand, due to greater sinkage. In addition, with the tail mounting point being at the robot's most rear, when the robot is at an angle its tail becomes even closer to the medium. Tail drag was also observed to be intermittent due to the oscillating/yawing motion of the robot. However, it should be noted that trials with continuous tail drag were not considered. Five different wheg designs from 3 to 7 spokes were used to study the effect of nominal stride length (Figure 4). Nominal stride length is the distance between each consecutive spoke wheg contact upon solid (nondeformable) surface (without experiencing any aerial phase or sinkage), while stride length is the measured footprint within the dry and wet medium (i.e., 0%, 15%, and 30% saturation) left behind by the consecutive spoked whegs. Eight unique motor frequencies varying from 2.33 to 7.43 Hz were chosen to generate a range of stride frequencies. The heterogeneity in form and composition of dry sand introduces uneven reaction from the medium. This would cause the whegs to momentarily become in phase and/or cause arced travel trajectories at higher motor frequencies with lower spoked whegs, both of which were minimally observed. Therefore, only straight robot trajectories (i.e., no arcs or side collisions with bed walls) were considered as successful runs, with a total of three trials per parameter condition. This would yield a total of 360 trials per each trialed gait (i.e., bipedal and quadrupedal). However, due to failures under certain parameter conditions, there ended up being a total of 294 bipedal trials and 343 quadrupedal trials. For example, the lowest motor frequency and spoke numbered wheg design would not provide sufficient thrust for the robot to move forward in dry sand. After each test run, the stride length (i.e., wheg tracks made in the medium) was measured manually using a digital caliper at three regions along the wheg tracks, and its mean was calculated. Stride

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TABLE 1 Definition of used parameters

Parameters	Symbol	Units
Stride frequency	f	1/s
Stride length	λ	m
Leg length	l_0	m
Velocity	v	m/s
Body mass	m	kg
Gravity	g	m/s^2
Electrical power	P	W
Motor efficiency	η	

TABLE 2 Equations used for normalization

Parameters	Symbol	Equation
Normalized stride frequency	$\bar{f}$	$f\sqrt{l_0/g}$
Normalized stride length	$\bar{\lambda}$	λ/l_0
Normalized velocity	$\bar{v}$	$v/\sqrt{l_0 g}$
Cost of transport	COT/η	$P/(mgv\eta)$

frequency (i.e., number of strides taken in unit time) and electrical current were synchronously measured by analyzing high-speed side view video recordings. The stride frequency was measured by counting the number of wheg spoke contact per second, and the robot's input current was recorded from the tethered (in frame) DC power supply. With the above-obtained information along with the measured electrical current, the COT (i.e., input power divided by the robot's weight and velocity) was calculated for each parameter condition.

2.5 | Nondimensionalization of experimental results

To analyze the robot's performance and efficiency, the effect of the running gait, medium's saturation level, motor frequency (directly proportional to stride frequency), and wheg spoke number (inversely proportional to nominal stride length) upon the robot's velocity and COT was evaluated (Table 1). The robot's body velocity was calculated by multiplying its stride length by stride frequency. Using the robot's body mass (m) and wheg outer contact radius (l_0), velocity (v), stride length (λ), and stride frequency (f) were normalized (Hof, 1996). The robot's COT was calculated with the input power: $P = IV$, where I is current and V is voltage (Table 2).

The WhegRunner's eight motor speeds were evenly spaced between 2.33 and 7.43 Hz. For this frequency range, Pololu's (50:1 micro-DC) geared motor performance chart (Pololu Corporation, 2018) shows that the motor efficiency increases as motor frequency increases, with the maximum tested motor speed of 7.43 Hz nearing the motor's maximum efficiency. To take into account the varying motor efficiency, the COT values were divided by their associated motor efficiency (COT/η). This was done by using the motor performance chart to obtain motor torque values per the motor frequency, and then using motor torque values to determine the motor efficiencies. Under loading conditions, the motors could not operate past 7.43 Hz without being damaged and thus motor speed could not be tested at a higher frequency.

2.6 | Statistical analysis

The data exhibited normal distribution trends and had residuals of equal variance. Therefore, a two-way multivariate analysis of variance (MANOVA; using IBM SPSS Statics 25) was conducted on the normalized data, evaluating the following dependent variables: normalized stride length, normalized velocity, and COT/η. Gait type, saturation level, wheg design, and motor frequency were set as the predictors. A curvilinear trend was observed from the COT/η plot (Figure 5), thus logarithm (base 10) transformation was used to linearize it, suitable for the analysis. Post hoc test was performed for pairwise comparisons between saturation level, wheg design, and motor frequency. Statistical analysis was performed per gait and between the gaits. Dependent variables were shown to be greatly influenced by all predictors. The terminology of stride frequency (as predictor) will be used as opposed to the motor frequency when presenting results, as they are directly proportional to one another.

3 | RESULTS AND DISCUSSION

Similar trends were observed between the bipedal and quadrupedal gaits, thus they will be presented together. The results will be presented in the following order: stride length, body forward velocity, and COT. Lastly, the performance of the two gaits will be compared.

3.1 | Stride length

With body forward velocity being the product of stride length and stride frequency, the effect of a number of spoked whegs (i.e., nominal stride length) and the visualization of the normalized stride length versus normalized stride frequency results will assist the comprehension of the body forward velocity results. When normalized stride length was plotted with respect to normalized stride frequency (Figure 6) for each wheg design (3–7-spoked whegs) and saturation level (0%–30%), the following dominant trends were observed: (1) Stride length increased as the number of spokes decreased (MANOVA, Bipedal: $F_{4,267} = 555.532$, $p \leq 0.00001$; Quadrupedal: $F_{4,315} = 273.811$, $p \leq 0.00001$). (2) For dry sand (0% saturation), stride length decreased as stride frequency increased (MANOVA, Bipedal: $F_{7,267} = 12.740$, $p \leq 0.00001$; Quadrupedal: $F_{7,315} = 2.661$, $p = 0.011$). (3) For 15% and 30% saturation, stride

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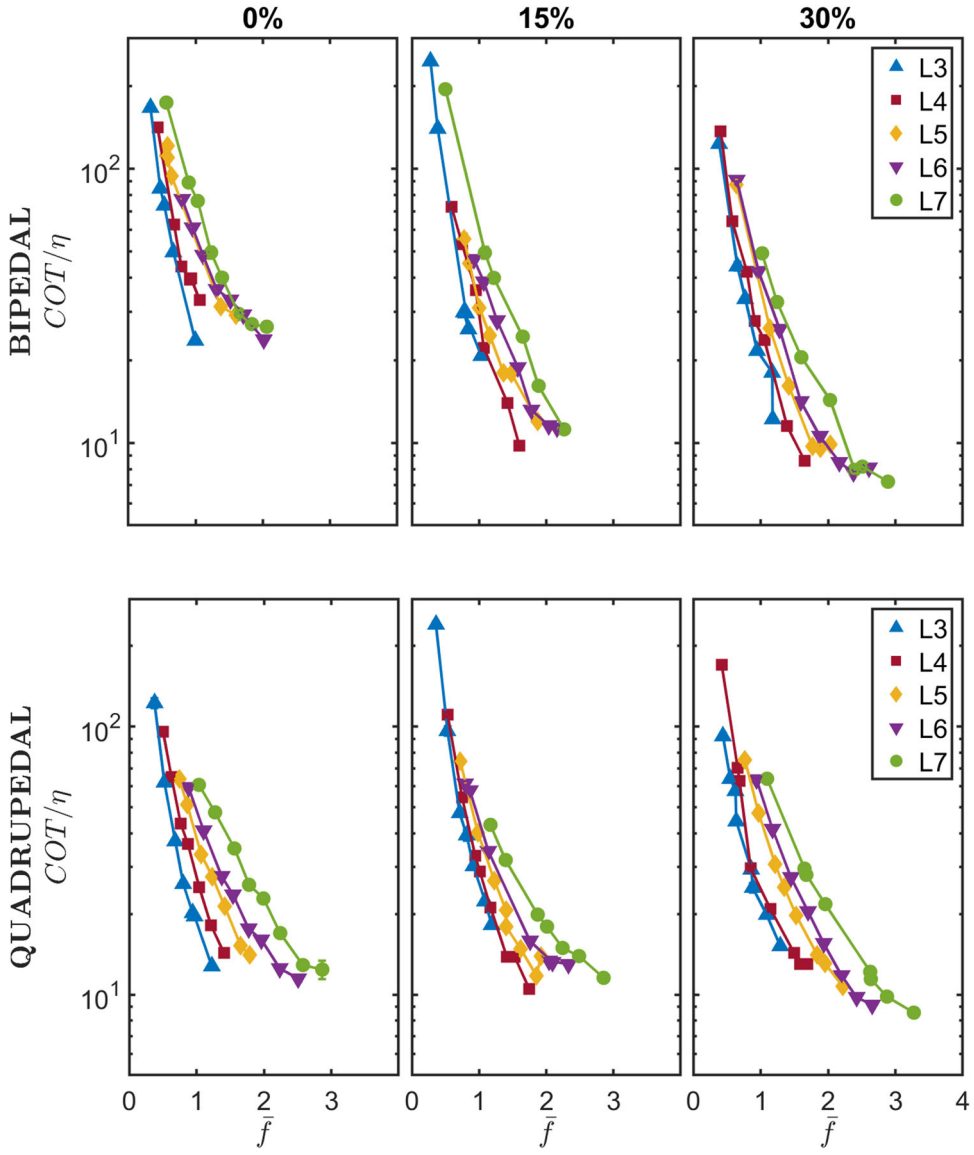


FIGURE 5 Robot cost of transport (COT/η) versus normalized stride frequency ($\bar{f}$) at various saturation levels using different wheg designs. Three trials were performed per parameter condition; that is, saturation, wheg design, and motor frequency.

length increased or remained stable as stride frequency increased (MANOVA, Bipedal: $F_{6,267} = 9.127$, $p \leq 0.00001$ and $F_{7,267} = 9.966$, $p \leq 0.00001$; Quadrupedal: $F_{7,315} = 3.314$, $p = 0.002$ and $F_{7,315} = 5.935$, $p \leq 0.00001$). (4) Stride length increased as saturation increased from 0% to 30% (MANOVA, Bipedal: $F_{2,267} = 210.447$, $p \leq 0.00001$; Quadrupedal: $F_{2,315} = 52.148$, $p \leq 0.00001$).

The whegs were purposely designed as rigid bodies, to focus on stride length and medium's reaction, averting from the effects of compliance on body velocity and energetics. Therefore, the observed general trend of stride length depends on the gap distance between spokes, which was deliberately created by the wheg geometry. In particular, nominal stride length is inversely proportional to wheg spoke number: as spoke number increases, the gap between each spoke decreases, and thus nominal stride length decreases. At 0% saturation, stride length was observed to decrease with the increase of stride frequency. Dry mediums like that of sand act like rigid particles, which easily "flow" under the applied stress (Grujicic, Marvi, Arakere, Bell, et al., 2010; Grujicic, Marvi, Arakere, & Haque, 2010; Li et al., 2013; Richefeu et al., 2009). If the motor frequency is too high, slippage occurs between the whegs and the medium, causing the robot to swim in its place, digging itself into the medium (see supplementary video; Li et al., 2009, 2013). In contrast, at 15% and 30% saturation, stride length increased as frequency increased. With the presence of liquid bonds (Mitarai & Nori, 2006) the medium becomes more cohesive, resulting in increased strength (Richefeu et al., 2009). Induced by the medium, this allows for the increase of push-off force. As the stride frequency increased, the aerial phase became more pronounced, especially with lower wheg spoke numbers (see supplementary video). The stride length increased as saturation increased from 0% to 15% to 30%. This indicates that as

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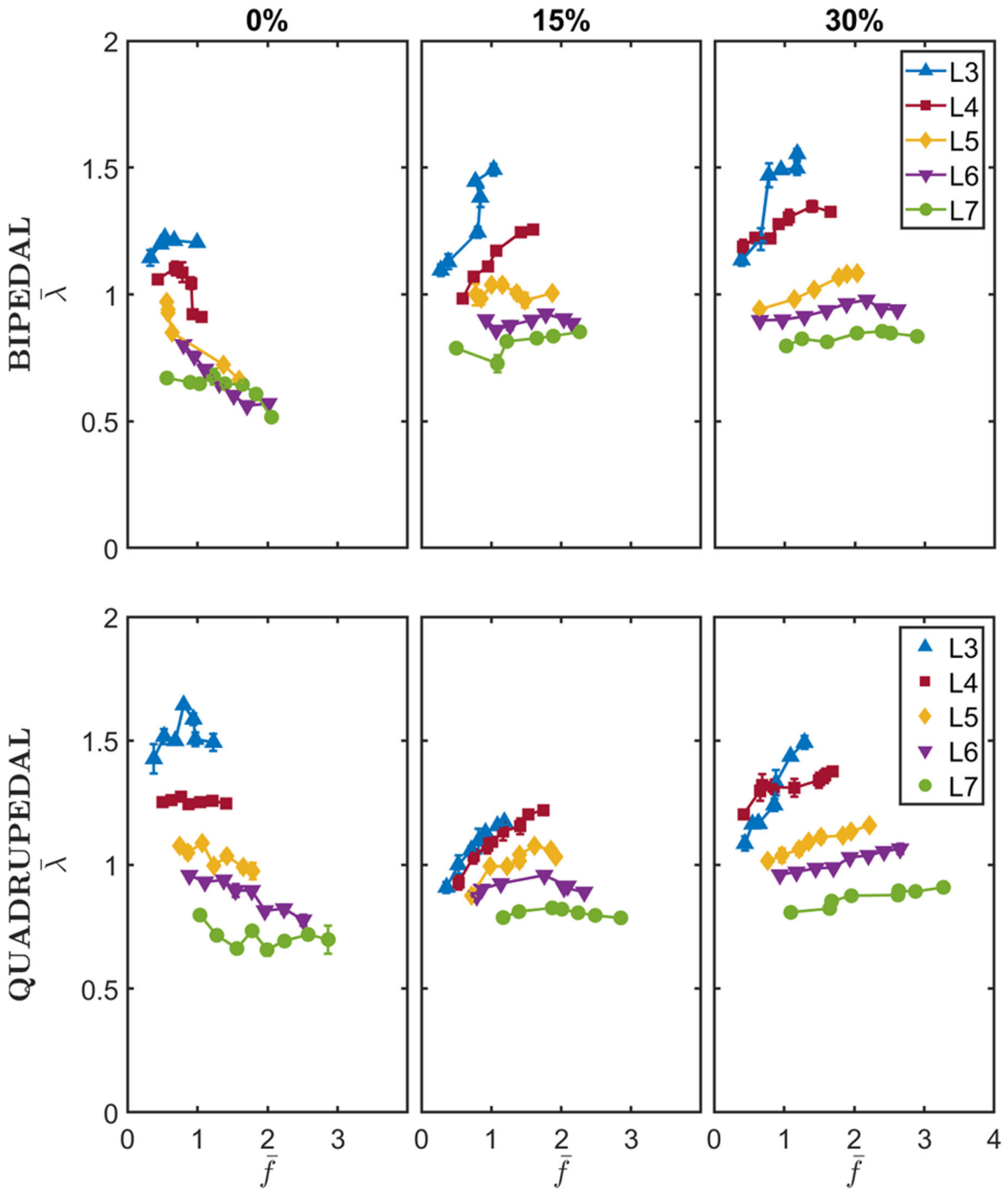


FIGURE 6 Robot normalized stride length ($\bar{\lambda}$) versus normalized stride frequency ($\bar{f}$) at various saturation levels using different wheg designs. Three trials were performed per parameter condition; that is, saturation, wheg design, and motor frequency

the cohesiveness of the medium increased, the occurrence of sinkage and slippage decreased, while the aerial phase increased.

3.2 | Body forward velocity

Examining the normalized velocity versus normalized stride frequency plot in Figure 7, the following conclusions could be made: (1) As stride frequency increased, the velocity increased (MANOVA, Bipedal: $F_{7,267} = 265.118$, $p \leq 0.00001$; Quadrupedal: $F_{7,315} = 323.347$, $p \leq 0.00001$). (2) Per any given stride frequency, as the wheg spoke number decreased, the velocity increased. (3) As saturation increased, the velocity increased (MANOVA, Bipedal: $F_{2,267} = 405.635$, $p \leq 0.00001$; Quadrupedal: $F_{2,315} = 105.798$, $p \leq 0.00001$).

Body velocity can either be altered through stride frequency and/or stride length. With body velocity being the product of stride

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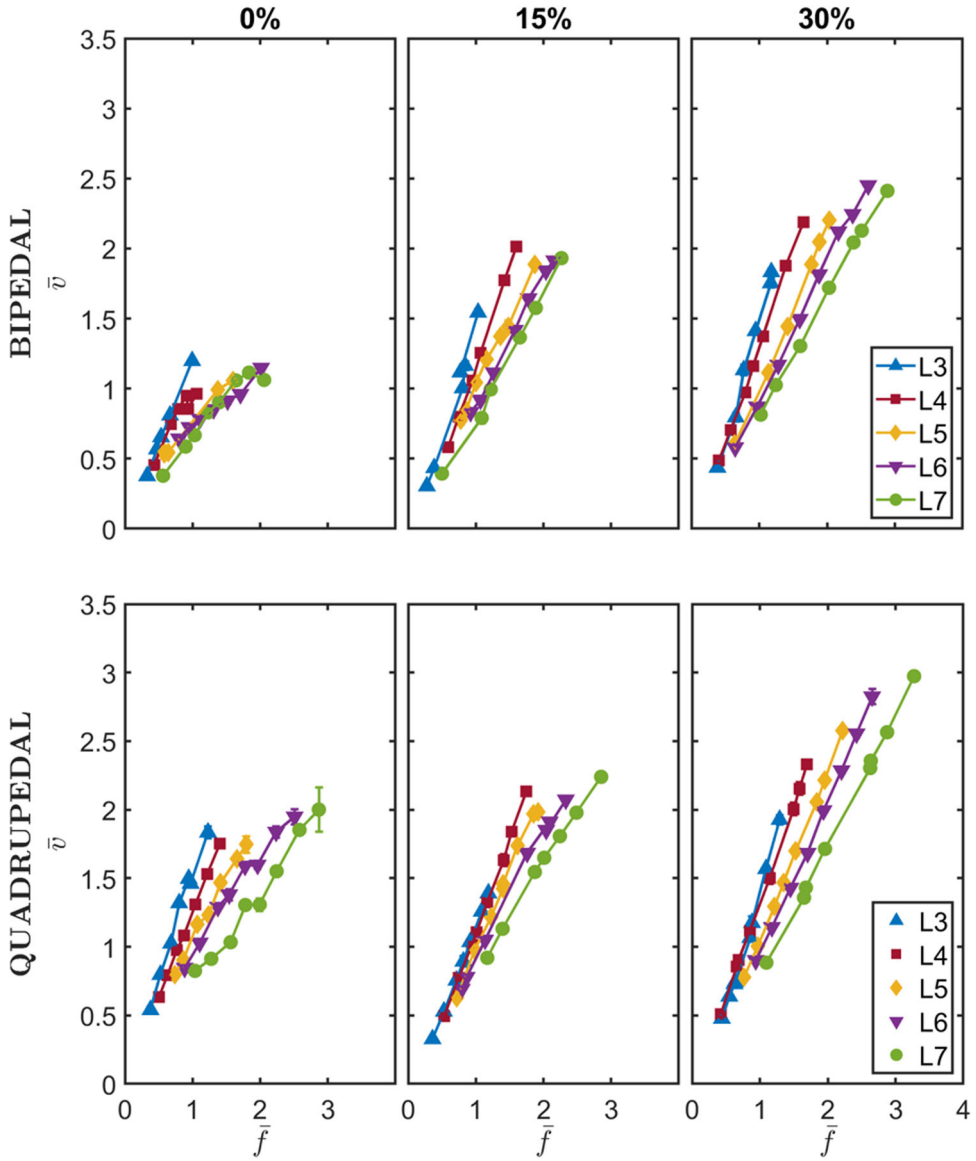


FIGURE 7 Robot normalized velocity ($\bar{v}$) versus normalized stride frequency ($\bar{f}$) at various saturation levels using different wheg designs. Three trials were performed per parameter condition; that is, saturation, wheg design, and motor frequency.

length by stride frequency, if stride length is kept constant, velocity would rise by the increase in stride frequency. This is what has been observed in the velocity plots. Similar experimental and simulation (via resistive force model) trends have been observed and presented in Li et al. (2013), where a six c-legged robot was tested up to 5 Hz stride frequency (via alternating tripod gait) upon dry granular media (i.e., poppy seed). The two curve-legged runners (circular track) test apparatus presented in Wu et al. (2016) showed a similar trend when running on chia seed for stride frequency ranging from 6 to 16 Hz. With respect to saturated medium, Liang et al. (2012) examined their two curve-legged runners (linear track) test apparatus on 23%, 27%, and 29% saturated "farm land" (lacking leafs and stones). While details of the medium's particle size are unknown and the highest tested leg angular velocity is 6 rad/s (stride frequency of 1 Hz), similar general trends were observed for their leg III (analogous to the c-shaped leg). This was observed to be associated with higher leg propulsion forces as the leg rotational speed increased, especially at 23% saturation.

When stride frequency is conserved, velocity is increased by the increase of nominal stride length, which is inversely proportional to wheg spoke number. As observed, the decrease in wheg spoke number resulted in the increase in body velocity. To the best of our knowledge, the effect of nominal stride length has not been systematically studied upon granular media like that presented here. With taxel (i.e., TActile piXEL) pressure sensors covering the outer perimeter of their two curve-legged runners, Wu et al. (2016) observed the contact force in granular media to be greatest at the outer proximal perimeter of the legs, closer to the point of rotation.

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At this phase the leg penetrates, compacts, and pushes off of the medium. It can be deduced that with a higher number of spoked whegs, less (surface area) contact is established between each leg and the medium, thus decreasing penetration and push-off force. Lower spoked whegs would be advantageous, as they introduce greater contact area, resulting in greater push-off force.

To determine if stride length or stride frequency is the dominant factor for the increase of body forward velocity, a MANOVA was performed, which showed both stride length (MANOVA, Bipedal: $F_{4,196} = 2189.495$, $p \leq 0.00001$; Quadrupedal: $F_{4,228} = 1854.423$, $p \leq 0.00001$) and stride frequency (MANOVA, Bipedal: $F_{7,196} = 6802.162$, $p \leq 0.00001$; Quadrupedal: $F_{7,228} = 5408.574$, $p \leq 0.00001$) are statistically significant, regardless of medium saturation.

From the significant increase in body forward velocity between 0% and 30% saturated sand, it can be deciphered that the cohesion of the medium decreases the events of sinkage and slippage, while increases the push-off force and occurrence of the aerial phase. With aerial phases being more prominent with lower spoked whegs, it indicates that wider stride length may be more impactful in attaining higher body velocities on the wet sand. Therefore, as saturation increased, stride length increased, and thus body velocity increased. The study conducted by Liang et al. (2012) presents adverse effects: as saturation increased from 23% to 29%, forward velocity decreased for their two curved-legged testing apparatus. It was observed that leg propulsion force and efficiency decreased with the increase of medium saturation. These conflicting results convey the important effect of the medium's particle size and stiffness/compaction at the saturated and dry state, which neither have been reported in the study and thus an impartial comparison cannot be made. It has been observed that depending on medium composition, its stiffness will either increase or decrease with the increase of saturation (Mitarai & Nori, 2006). Li et al. (2009) showed that as dry granular medium (poppy seed) increased in volume fraction (0.580 to 0.633), body forward velocity also increased. For 520 μm dry glass beads, Marston et al. (2012) showed that penetration depth decreased with the increase in volume fraction (0.59–0.63). However, penetration depth increased with the increase of impact velocity. This indicates that in dry granular media, higher volume fraction and lower stride frequencies will provide a greater push-off force and lesser occurrence of sinkage and slippage. While in saturated mediums within the funicular state, the increase in saturation will increase the medium's yield strength, and thus the robot's push-off force.

3.3 | Cost of transport

To evaluate the energetics of robot locomotion, mechanical COT was evaluated (Figure 5). The plot for COT over motor efficiency (COT/η) versus stride frequency exhibited the following patterns: (1) As frequency increased, the COT/η decreased (MANOVA, Bipedal: $F_{7,267} = 167.299$, $p \leq 0.00001$; Quadrupedal: $F_{7,315} = 378.577$, $p \leq 0.00001$). (2) Per any given stride frequency, as wheg spoke number decreased, COT/η decreased. (3) Overall, as saturation increased, COT/η decreased (MANOVA, Bipedal: $F_{2,267} = 214.069$, $p \leq 0.00001$; Quadrupedal: $F_{2,315} = 8.916$, $p = 0.0002$).

COT/η decreased with the increase of frequency, since the wheg spoke contact time decreased, reducing the event of sinkage and drag. With less sinkage, less torque and thus current (at a constant voltage of 12) is required to push the robot forward. Furthermore, the robot's center of mass experiences less vertical oscillations at higher frequencies, resulting in less energy dissipation. The two curve-legged runner test apparatus by Liang et al. (2012), which was tested upon 23%, 27%, and 29% saturated "farm land" showed opposite trends. They presented leg propulsive efficiency (i.e., distance traveled divided by energy input) decreasing with the increase in leg angular velocity. With propulsion (forward) efficiency decreasing, the COT would be increasing. Once again, since the details of the medium (i.e., particle size and stiffness) are not provided, the results cannot be compared. Per a given stride frequency, COT/η decreased with the wheg spoke number decreasing. The observation can be understood by reflecting upon the equation for the cost of transport ($COT = P/(mgv)$). As it was stated earlier, velocity increased as the wheg spoke number decreased per any given stride frequency. Therefore, the higher velocity achieved by lower spoke numbers will produce lower COT/η at a given stride frequency. Finally, the overall trend dictates that as saturation increases COT/η decreases, which is rooted upon the same concepts. As the sand shifts from a deformable to more cohesive medium, sinkage, and slippage decrease, requiring less power to push forward.

3.4 | Bipedal versus quadrupedal

When comparing the quadrupedal in respect to bipedal experimental results, the following additional observations could be made: quadrupedal gait generally achieves (1) higher body velocities (MANOVA, $F_{1,631} = 36.205$, $p \leq 0.00001$), (2) wider strides (MANOVA, $F_{1,631} = 17.354$, $p = 0.00004$), and (3) lower COT (MANOVA, $F_{1,631} = 11.013$, $p = 0.001$). As opposed to the bipedal gait, which theoretically has one point of contact per given time, the quadrupedal gait introduced a minimum of two (diagonal) points of contact at a given time (except during the aerial phase). Hence, the robot's contact forces are distributed over its balanced strides, increasing its stability during mobility. This was shown to be most beneficial in dry sand, where the quadrupedal gait attains wider strides and thus higher body velocity as stride frequency increased. For saturated sand, by decreasing localized sinkage upon a wheg, such as in the case of bipedal gait, the robot is capable of increasing its velocity as the stride frequency increases. In addition, with the quadrupedal gait providing a more stable locomotion (i.e., more uniform distribution of contact forces) and not having to stabilize body pitch (as needed during the bipedal gait), less effort for stabilization was required to traverse over the mediums.

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4 | CONCLUSION

WhegRunner, a bipedal/quadrupedal locomotor with whegs was developed and used as a ground mobility platform to examine how stride length, stride frequency, ground saturation level, and running gait affect the robot's body forward velocity and COT. Similar trends were observed between the bipedal and quadrupedal gait. It was observed that as the wheg spoke number decreased, the robot's stride length increased. This is because the nominal stride length is inversely proportional with wheg spoke number. Wider strides lead to higher body forward velocity. As the sand saturation level increased from 0% to 15% to 30%, the stride length and forward velocity predominantly increased. With the increase of liquid bonds within the medium, the cohesion of the medium increases, increasing its internal yield strength. This allows for greater wheg traction and push-off force, and less sinkage and slippage. Body forward velocity is a product of stride length and stride frequency, therefore, as both parameters increase, so does the velocity. In addition, per a given stride frequency, velocity increased with the increase of stride length (i.e., decrease of wheg spoke number). With the increase in stride frequency and saturation level, the COT decreased. The decreasing occurrence of sinkage and slippage leads to less required torque to propel the robot forward. Furthermore, with the robot's center of mass experiencing less vertical oscillation, the energy dissipation was reduced. Like body forward velocity, per a given stride frequency, the lowest COT was attained with lower wheg spoke number (i.e., higher stride length). This can be understood by reflecting on the definition of COT, where wider stride lengths lead to the increase of velocity and thus the decrease of COT. Lastly, when comparing the robot's quadrupedal to bipedal gait performance, quadrupedal gait showed to be more energy efficient and achieved higher forward velocity. This was accomplished through greater distribution of applied force and stabilized forward mobility. The obtained knowledge can be channeled through the development of efficient and effective exploration robots for diversified terrains and mediums.

For future directions, it would be beneficial to evaluate higher motor frequencies with greater capacity motors, to determine if an optimum will be achieved. It would also be worthwhile to investigate different increments of saturation, to see how the trends develop. Lastly, through coupling multibody dynamic simulation (i.e., robot dynamics) with discrete element method (i.e., granular media modeling), mobility on granular media can be further studied computationally. For instance, the impact of particle size, geometry, volume fraction, cohesion, and wheg designs on robot performance can be explored in a systematic manner to seek optimal designs.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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REFERENCES

Brown, C.Y., Vogtmann, D.E. & Bergbreiter, S. (2013) Efficiency and effectiveness analysis of a new direct drive miniature quadruped robot. *2013 IEEE international conference on robotics and automation*. Piscataway, New Jersey, USA: IEEE, pp. 5631–5637.

Bruzzone, L. & Quaglia, G. (2012) Locomotion systems for ground mobile robots in unstructured environments. *Mechanical Sciences*, *3*(2), 49–62.

Crespi, A., Badertscher, A., Guignard, A. & Ijspeert, A.J. (2005) Swimming and crawling with an amphibious snake robot. *Proceedings of the IEEE international conference on robotics and automation*. Piscataway, New Jersey, USA: IEEE, pp. 3024–3028.

Crespi, A. & Ijspeert, A.J. (2006) AmphiBot II: an amphibious snake robot that crawls and swims using a central pattern generator. *Proceedings of the 9th international conference on climbing and walking robots (CLAWAR)*, number BIOROB-CONF-2006-001, Piscataway, New Jersey, USA, pp. 19–27.

Crespi, A., Karakasiliotis, K., Guignard, A. & Ijspeert, A.J. (2013) Salamandra Robotica II: an amphibious robot to study salamander-like swimming and walking gaits. *IEEE Transactions on Robotics*, *29*(2), 308–320.

Eich, M., Grimminger, F. & Kirchner, F. (2008) A versatile stair-climbing robot for search and rescue applications. *2008 IEEE international workshop on safety, security and rescue robotics*. Piscataway, New Jersey, USA: IEEE, pp. 35–40.

Eich, M., Grimminger, F. & Kirchner, F. (2009) Proprioceptive control of a hybrid legged-wheeled robot. *2008 IEEE international conference on robotics and biomimetics*. Piscataway, New Jersey, USA: IEEE, pp. 774–779.

Floyd, S., Adilak, S., Ramirez, S., Rogman, R. & Sitti, M. (2008) Performance of different foot designs for a water running robot. *IEEE international conference on robotics and automation*. Piscataway, New Jersey, USA: IEEE, pp. 244–250.

Floyd, S., Keegan, T., Palmisano, J. & Sitti, M. (2006) A novel water running robot inspired by basilisk lizards. *IEEE/RSJ international conference on intelligent robots and systems*. Piscataway, New Jersey, USA: IEEE, pp. 5430–5436.

Floyd, S. & Sitti, M. (2008) Design and development of the lifting and propulsion mechanism for a biologically inspired water runner robot. *IEEE Transactions on Robotics*, *24*(3), 698–709.

Grujicic, M., Marvi, H., Arakere, G., Bell, W. & Haque, I. (2010) The effect of up-armoring of the high-mobility multi-purpose wheeled vehicle (HMMWV) on the off-road vehicle performance. *Multidiscipline Modeling in Materials and Structures*, *6*(2), 229–256.

Grujicic, M., Marvi, H., Arakere, G. & Haque, I. (2010) A finite element analysis of pneumatic-tire/sand interactions during off-road vehicle travel. *Multidiscipline Modeling in Materials and Structures*, *6*(2), 284–308.

Herbert, S.D., Drenner, A. & Papanikolopoulos, N. (2008) Loper: a quadruped-hybrid stair climbing robot. *2008 IEEE international conference on robotics and automation*. Piscataway, New Jersey, USA: IEEE, pp. 799–804.

Hof, A. (1996) Scaling gait data to body size. *Gait & Posture*, *3*(4), 222–223.

15564967, 2023, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/rob.22121 by Universita Degli Studi Di Genova, Wiley Online Library on [08/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Holm, P., Schaefer, T. & Kristensen, H. (1985) Granulation in high-speed mixers Part V. Power consumption and temperature changes during granulation. *Powder Technology*, *43*(3), 213–223.

Jun, B.-H., Shim, H. & Kim, B. (2013) Hexapod robot crabster CR200 for high tide, turbid water exploration six-legged robot built for stability, mobility in harsh environments. *Sea Technology*, *54*(10), 33.

Karakasiliotis, K., Thandiackal, R., Melo, K., Horvat, T., Mahabadi, N., Tsitkov, S., Cabelguen, J. & Ijspeert, A. (2016) From cineradiography to biorobots: an approach for designing robots to emulate and study animal locomotion. *Journal of the Royal Society Interface*, *13*(119), 20151089.

Kim, H., Lee, D., Liu, Y., Jeong, K. & Seo, T. (2016) Hexapedal robotic platform for amphibious locomotion on ground and water surface. *Journal of Bionic Engineering*, *13*(1), 39–47.

Kristensen, H., Holm, P. & Schaefer, T. (1985) Mechanical properties of moist agglomerates in relation to granulation mechanisms Part I. Deformability of moist, densified agglomerates. *Powder Technology*, *44*(3), 227–237.

Lambrecht, B.G., Horchler, A.D. & Quinn, R.D. (2005) A small, insect-inspired robot that runs and jumps. *Proceedings of the 2005 IEEE international conference on robotics and automation*. Piscataway, New Jersey, USA: IEEE, pp. 1240–1245.

Lee, D.G., Kim, H., Jeong, K. & Seo, T. (2014) Development of a robotic platform for amphibious locomotion on ground and water surfaces. *The 11th international conference on ubiquitous robots and ambient intelligence (URAI)*. Piscataway, New Jersey, USA: IEEE, pp. 417–418.

Li, C., Umbanhowar, P.B., Komsuoglu, H., Koditschek, D.E. & Goldman, D.I. (2009) Sensitive dependence of the motion of a legged robot on granular media. *Proceedings of the National Academy of Sciences*, 106(9), 3029–3034.

Li, C., Zhang, T. & Goldman, D.I. (2013) A terradynamics of legged locomotion on granular media. *Science*, *339*(6126), 1408–1412.

Liang, X., Xu, M., Xu, L., Liu, P., Ren, X., Kong, Z., Yang, J. & Zhang, S. (2012) The AmphiHex: a novel amphibious robot with transformable leg-flipper composite propulsion mechanism. *IEEE/RSJ international conference on intelligent robots and systems (IROS)*. Piscataway, New Jersey, USA: IEEE, pp. 3667–3672.

Low, K.-H., Zhou, C., Ong, T. & Yu, J. (2007) Modular design and initial gait study of an amphibian robotic turtle. *IEEE international conference on robotics and biomimetics*. Piscataway, New Jersey, USA: IEEE, pp. 535–540.

Ma, X., Guo, S., Li, L., Ji, Y. & Wang, Y. (2014) The underwater motion simulation of a spherical amphibious robot. *IEEE international conference on mechatronics and automation (ICMA)*. Piscataway, New Jersey, USA: IEEE, pp. 145–149.

Machado, J.T. & Silva, M.F. (2006) An overview of legged robots. *International symposium on mathematical methods in engineering*. Ankara, Turkey: MME Press.

Marston, J., Vakarelski, I.U. & Thoroddsen, S.T. (2012) Sphere impact and penetration into wet sand. *Physical Review E*, *86*(2), 020301.

Marvi, H., Gong, C., Gravish, N., Astley, H., Travers, M., Hatton, R.L., Mendelson, J.R., Choset, H., Hu, D.L. & Goldman, D.I. (2014) Sidewinding with minimal slip: snake and robot ascent of sandy slopes. *Science*, *346*(6206), 224–229.

Mazouchova, N., Gravish, N., Savu, A. & Goldman, D.I. (2010) Utilization of granular solidification during terrestrial locomotion of hatchling sea turtles. *Biology Letters*, rsbl20091041.

Mazouchova, N., Umbanhowar, P.B. & Goldman, D.I. (2013) Flipper-driven terrestrial locomotion of a sea turtle-inspired robot. *Bioinspiration & Biomimetics*, *8*(2), 026007.

Mitarai, N. & Nori, F. (2006) Wet granular materials. *Advances in Physics*, *55*(1–2), 1–45.

Neumeyer, M. & Jones, B. (1965) The marsh screw amphibian. *Journal of Terramechanics*, *2*(4), 83–88.

Osiński, D. & Szykiedans, K. (2015) Small remotely operated screw-propelled vehicle. In: Szewczyk, R., Zieliński, C. & Kaliczyńska, M. (Eds.) *Progress in automation, robotics and measuring techniques*. Midtown Manhattan, New York City:Midtown Manhattan, New York City: Springer, pp. 191–200.

Park, H.S., Floyd, S. & Sitti, M. (2009) Dynamic modeling and analysis of pitch motion of a basilisk lizard inspired quadruped robot running on water. *IEEE international conference on robotics and automation*. Piscataway, New Jersey, USA: IEEE, pp. 2655–2660.

Park, H.S., Floyd, S. & Sitti, M. (2010) Roll and pitch motion analysis of a biologically inspired quadruped water runner robot. *The International Journal of Robotics Research*, *29*(10), 1281–1297.

Park, H.S. & Sitti, M. (2009) Compliant footpad design analysis for a bio-inspired quadruped amphibious robot. *IEEE/RSJ international conference on intelligent robots and systems*. Piscataway, New Jersey, USA:Piscataway, New Jersey, USA: IEEE, pp. 645–651.

Pololu Corporation. (2018) *Pololu micro metal gearmotors*. Available from: https://www.pololu.com/file/0J1487/pololu-micro-metal-gearmotors.pdf [Accessed 2018-10-02].

Reina, G., Ojeda, L., Milella, A. & Borenstein, J. (2006) Wheel slippage and sinkage detection for planetary rovers. *IEEE/ASME Transactions on Mechatronics*, *11*(2), 185–195.

Ren, X., Liang, X., Kong, Z., Xu, M., Xu, R. & Zhang, S. (2013) An experimental study on the locomotion performance of elliptic-curve leg in muddy terrain. *IEEE/ASME international conference on advanced intelligent mechatronics (AIM)*. Piscataway, New Jersey, USA: Piscataway, New Jersey, USA: IEEE, pp. 518–523.

Richefeu, V., El Youssoufi, M.S., Azéma, E. & Radjai, F. (2009) Force transmission in dry and wet granular media. *Powder Technology*, *190*(1–2), 258–263.

Schubert, H., Herrmann, W. & Rumpf, H. (1975) Deformation behaviour of agglomerates under tensile stress. *Powder Technology*, *11*(2), 121–131.

Spagna, J.C., Goldman, D.I., Lin, P.-C., Koditschek, D.E. & Full, R.J. (2007) Distributed mechanical feedback in arthropods and robots simplifies control of rapid running on challenging terrain. *Bioinspiration & Biomimetics*, *2*(1), 9.

Stager, A., Karydis, K. & Tanner, H.G. (2015) A passively sprawling miniature legged robot. *2015 IEEE international conference on robotics and automation (ICRA)*. Piscataway, New Jersey, USA: IEEE, pp. 3134–3139.

Stone, M., Barry, R., Bernstein, D., Pelc, M., Tsui, Y. & Schiffer, P. (2004) Local jamming via penetration of a granular medium. *Physical Review E*, *70*(4), 041301.

Sun, Y. & Ma, S. (2013) A versatile locomotion mechanism for amphibious robots: eccentric paddle mechanism. *Advanced Robotics*, *27*(8), 611–625.

Thoesen, A., McBryan, T. & Marvi, H. (2019) Helically-driven granular mobility and gravity-variant scaling relations. *RSC Advances*, *9*(22), 12572–12579.

Thoesen, A., Ramirez, S. & Marvi, H. (2018) Screw-powered propulsion in granular media: an experimental and computational study. *2018 IEEE international conference on robotics and automation (ICRA)*. Piscataway, New Jersey, USA: IEEE, pp. 1–6.

Thoesen, A., Ramirez, S. & Marvi, H. (2019) Screw-generated forces in granular media: experimental, computational, and analytical comparison. *AIChE Journal*, *65*(3), 894–903.

Walker, I.D. (2011) Continuum robot appendages for traversal of uneven terrain in in situ exploration. *2011 Aerospace conference*. Piscataway, New Jersey, USA: IEEE, pp. 1–8.

Wang, M., Sun, L. & Wang, Y. (2006) Dynamic modelling and optimized energy distribution of amphibian walking robot. *Proceedings of the IEEE international conference on mechatronics and automation*. Piscataway, New Jersey, USA: IEEE, pp. 634–638.

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Wu, X.A., Huh, T.M., Mukherjee, R. & Cutkosky, M. (2016) Integrated ground reaction force sensing and terrain classification for small legged robots. *IEEE Robotics and Automation Letters*, *1*(2), 1125–1132.

Xu, L., Liang, X., Xu, M., Liu, B. & Zhang, S. (2013) Interplay of theory and experiment in analysis of the advantage of the novel semi-elliptical leg moving on loose soil. *IEEE/ASME international conference on advanced intelligent mechatronics (AIM)*. Piscataway, New Jersey, USA: IEEE, pp. 26–31.

Yu, J., Ding, R., Yang, Q., Tan, M., Wang, W. & Zhang, J. (2012) On a bio-inspired amphibious robot capable of multimodal motion. *IEEE/ASME Transactions On Mechatronics*, *17*(5), 847–856.

Yu, J., Ding, R., Yang, Q., Tan, M. & Zhang, J. (2013) Amphibious pattern design of a robotic fish with wheel–propeller–fin mechanisms. *Journal of Field Robotics*, *30*(5), 702–716.

Yu, S., Ma, S., Li, B. & Wang, Y. (2011) An amphibious snake-like robot with terrestrial and aquatic gaits. *IEEE international conference on robotics and automation (ICRA)*. Piscataway, New Jersey, USA: IEEE, pp. 2960–2961.

Zarrouk, D., Pullin, A., Kohut, N. & Fearing, R.S. (2013) Star, a sprawl tuned autonomous robot. *2013 IEEE international conference on robotics and automation*. Piscataway, New Jersey, USA: IEEE, pp. 20–25.

Zarrouk, D. & Yehezkel, L. (2018) Rising star: a highly reconfigurable sprawl tuned robot. *IEEE Robotics and Automation Letters*, *3*(3), 1888–1895.

Zhang, S., Liang, X., Xu, L. & Xu, M. (2013) Initial development of a novel amphibious robot with transformable fin–leg composite propulsion mechanisms. *Journal of Bionic Engineering*, *10*(4), 434–445.

Zhong, B., Zhou, Y., Li, X., Xu, M. & Zhang, S. (2016) Locomotion performance of the amphibious robot on various terrains and underwater with flexible flipper legs. *Journal of Bionic Engineering*, *13*(4), 525–536.

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